

CSE 325: INFORMATION SYSTEM DESIGN

Week 1: Introduction

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CSE 325: INFORMATION SYSTEM DESIGN

What is an Information System?

A system that **collects** , **stores** , **processes** , and **distributes** data to provide valuable information to users

Examples of Information System

Banking apps & ATM systems

E-commerce platforms

University registration portals

Supply chain management systems

Hospital information systems

CRM tools (e.g., Salesforce, HubSpot)

Components of Information System

- People
 - Individuals who interact with, maintain, or benefit from the system
 - For example, end users, staff/operators, management, IT professionals

Components of Information System

- Processes
 - Rules, workflows, and procedures that define how tasks are performed
 - Includes
 - Step-by-step workflows
 - Business rules (e.g., payment must be verified before order dispatch)
 - Policies
 - Standard operating procedure

Components of Information System

- Technology
 - Tools that enable the IS to function
 - Includes
 - Hardware: devices, servers
 - Software: applications, OS, databases
 - Networks: Wi-Fi, mobile data, cloud infrastructure

Components of Information System

- Data
 - Raw material the system uses and transforms
 - Includes
 - Stored data (databases, logs, history)
 - Real-time input data (sensor readings, transactions)
 - Output data (reports, dashboards, notifications)

Components of Information System

- Organizational Context
 - **Environment** in which the system operates
 - Includes
 - Business goals & strategy
 - Culture and norms
 - Legal or regulatory requirements
 - Physical and social environment
 - Budget and resource constraints

Components of Information System

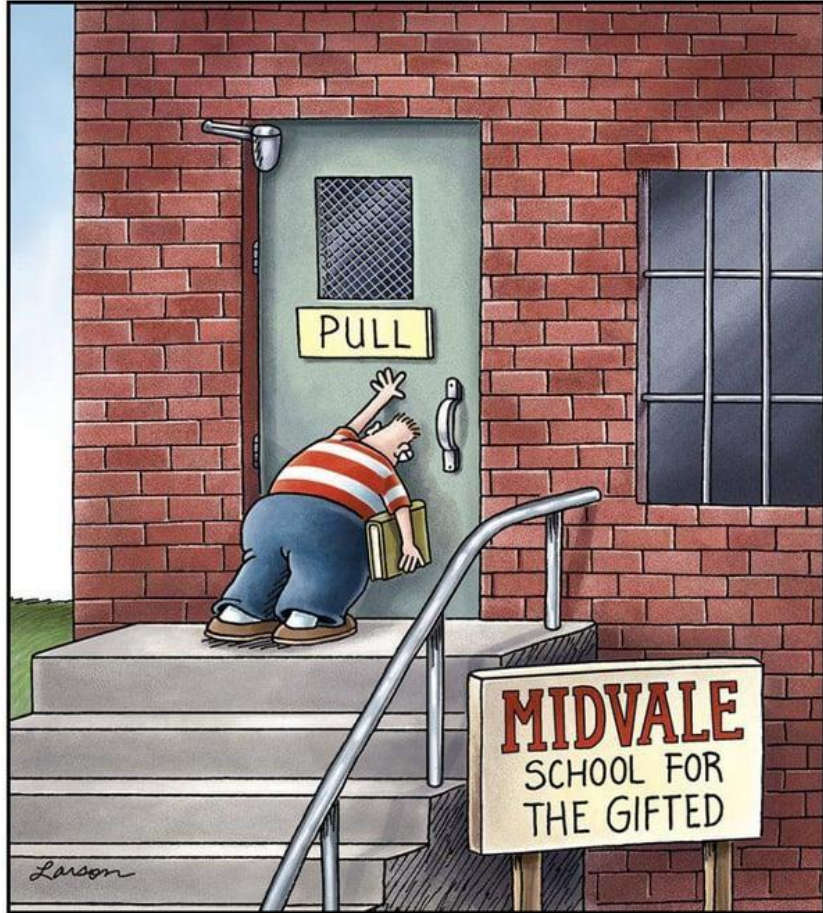
For an online food delivery system (FoodPanda, Uber Eats)

- People
- Processes
- Technology
- Data
- Organizational Contexts

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DOOR QUIZ





Door Quiz

Say it loud, what action do you use to open the door

- Push
- Pull

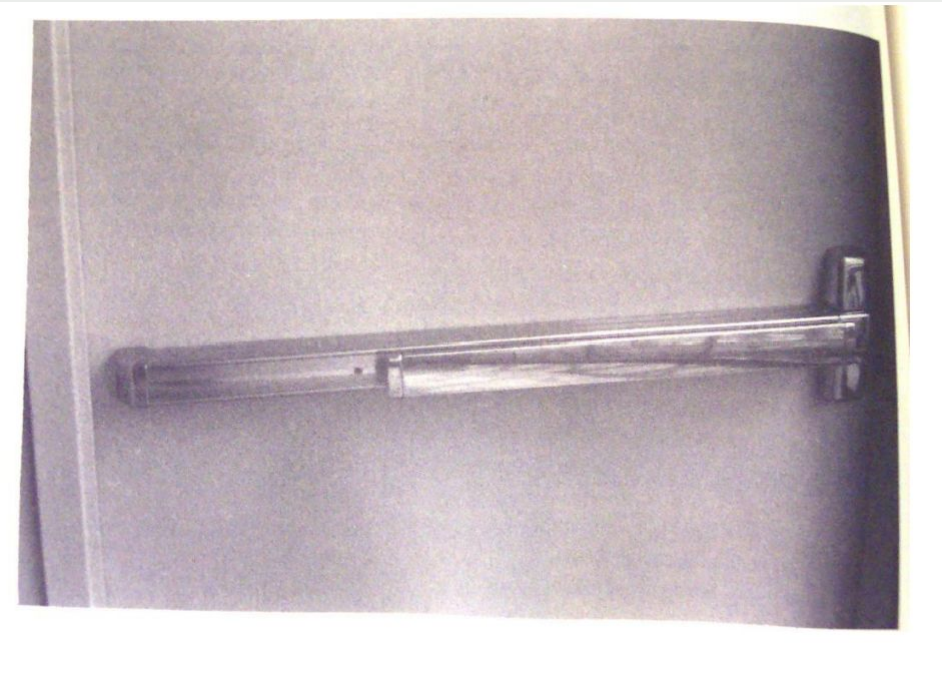


Door Quiz



Door Quiz

Door Quiz





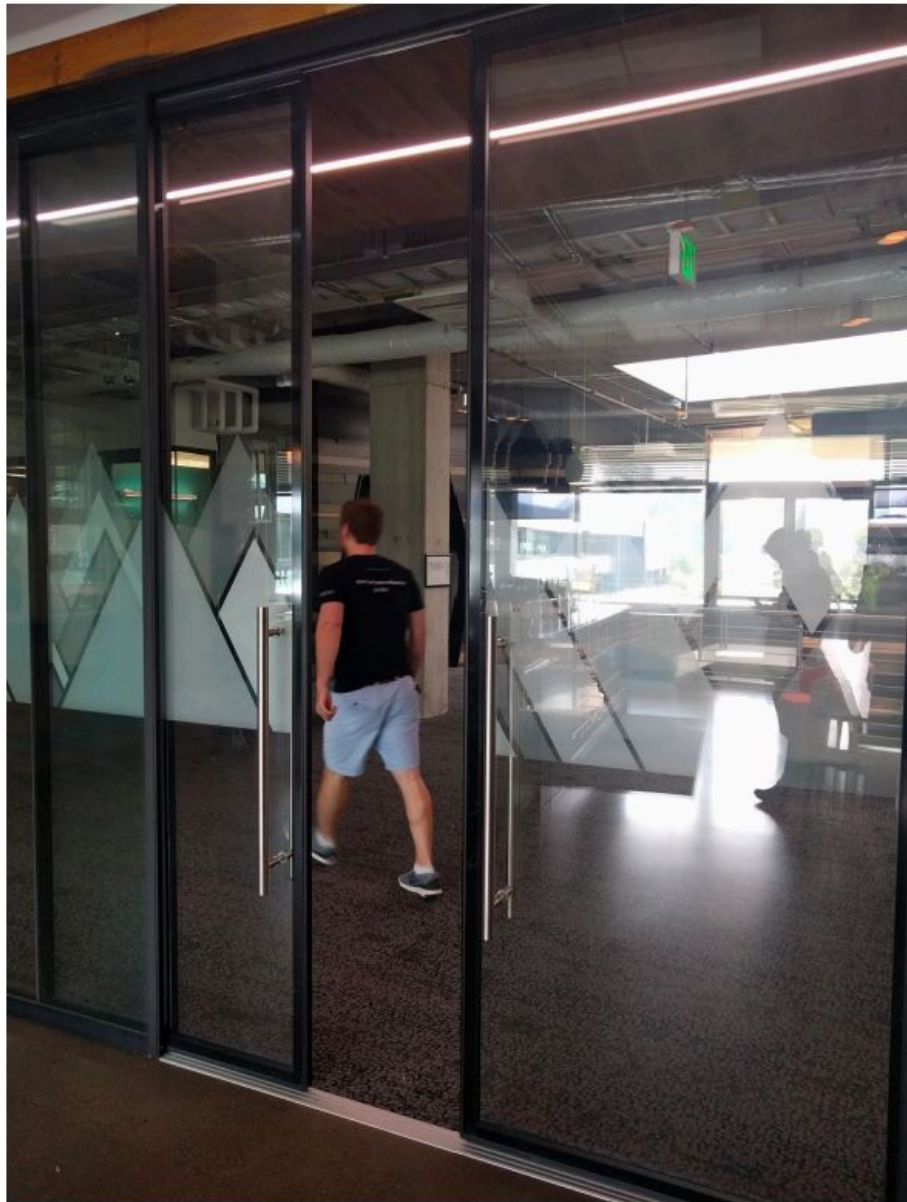
Door Quiz

Door Quiz

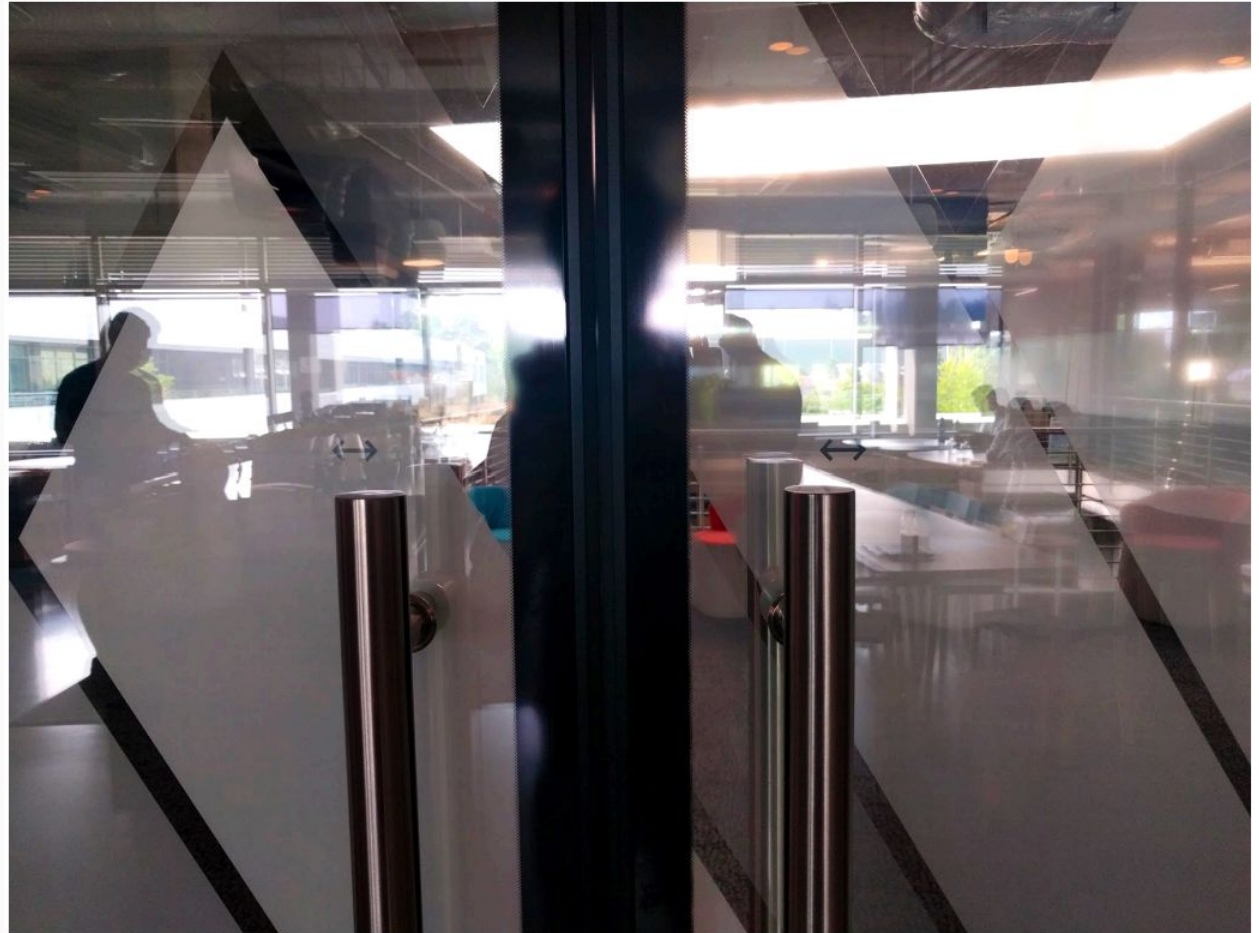


Door Quiz





Door Quiz



Signs don't Help



What is Special about IS?

- Nothing
- It's about good designs and bad designs
- We make push/pull decisions many times per day
- We all encounter doors that do this badly
- We all see signs that do not change what we do

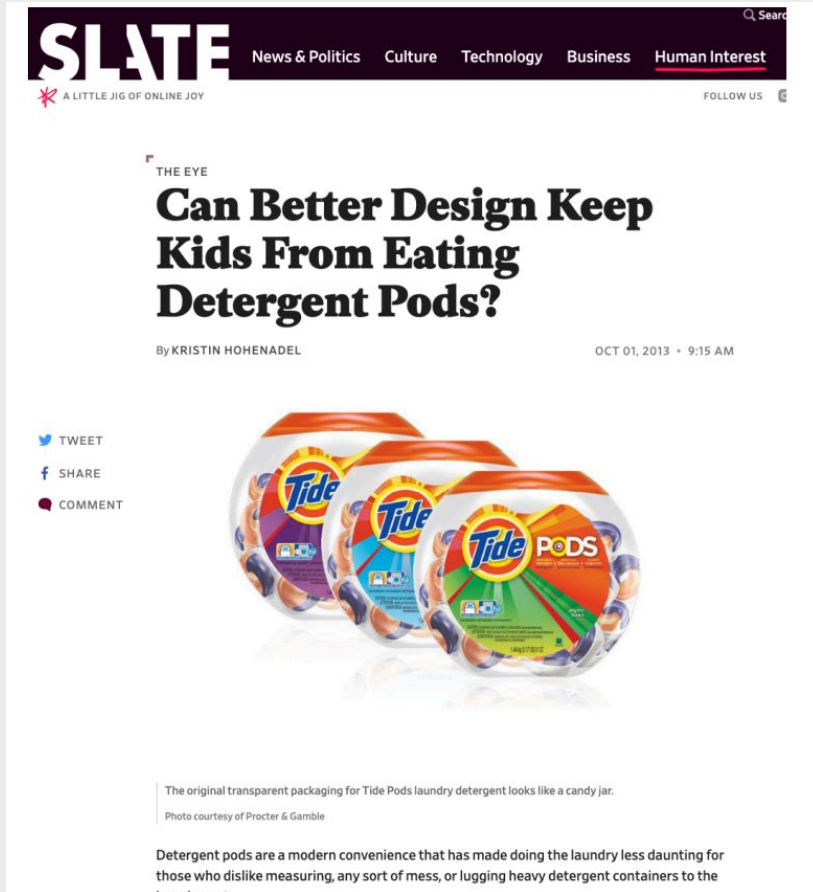
“Is a system only successful if it’s usable? What if it’s usable—but no one wants to use it?”



Think and Share Activity

“Think about a technology you used— a website, app, device, or system — that frustrated you. What went wrong, and how did you respond?” (5 mins)

Poor Design can have Serious Impacts



<https://asn.flightsafety.org/asndb/325614>

“Good Design” Means

- Systems are built for humans; must be designed for the user
- Recognize individual differences; appreciate design implications of these human factors
- Recognize the design of things, procedures, etc., influences human behavior and well-being
- Emphasize empirical data & evaluation
- Rely on the scientific method
- Things, procedures, environments, and people do not exist in isolation

Good Design Is Not...

☹ **NOT just applying checklists and guidelines**

- These can help, but UCD is a whole philosophy

☹ **NOT using oneself as the model user**

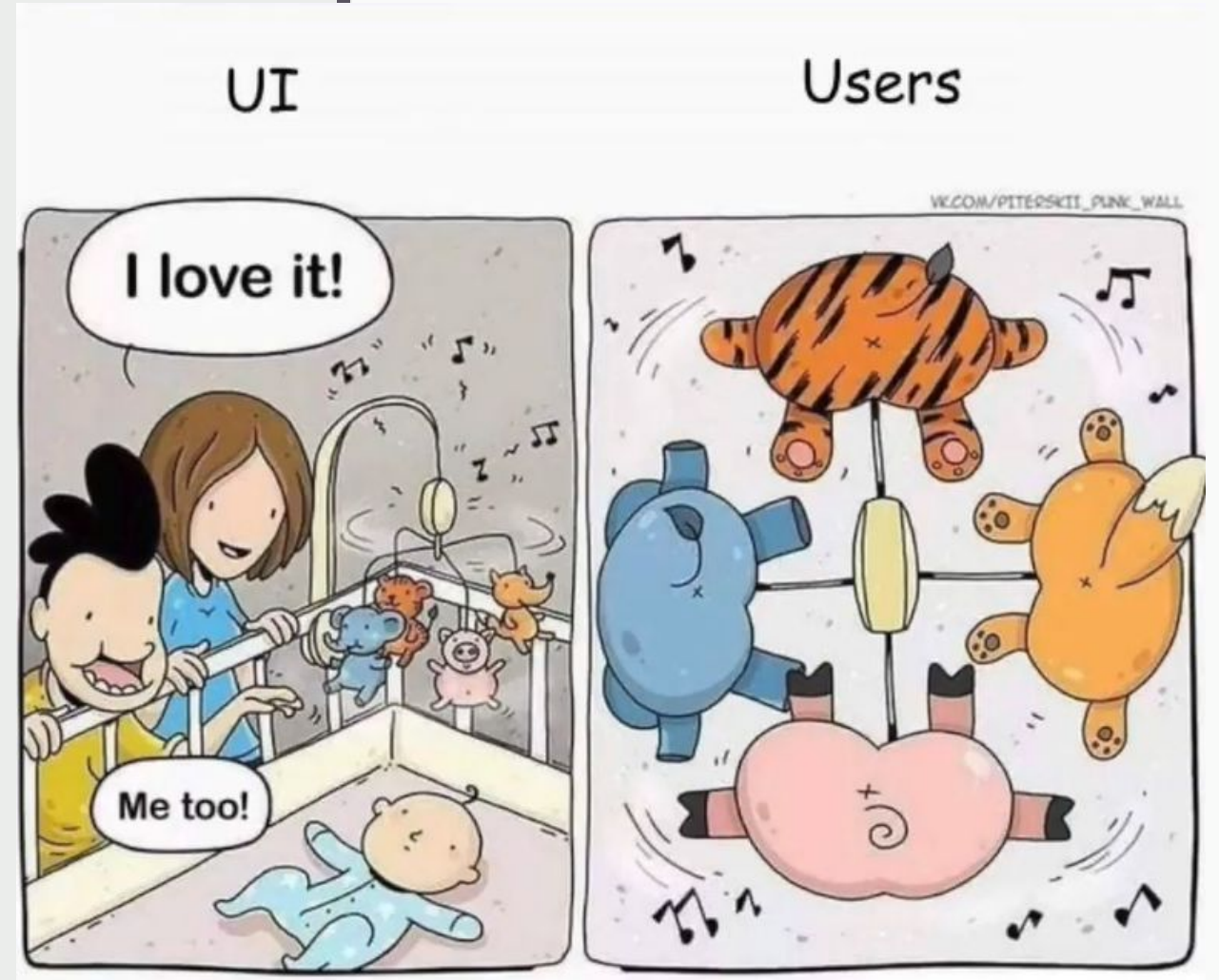
- Know your real users; recognize variation in humans

☹ **NOT just common sense**

- Knowing how to design a fire alarm so it will be heard over background noise is not something we all know
- The HF specialist knows where or how to get the information needed to answer design questions

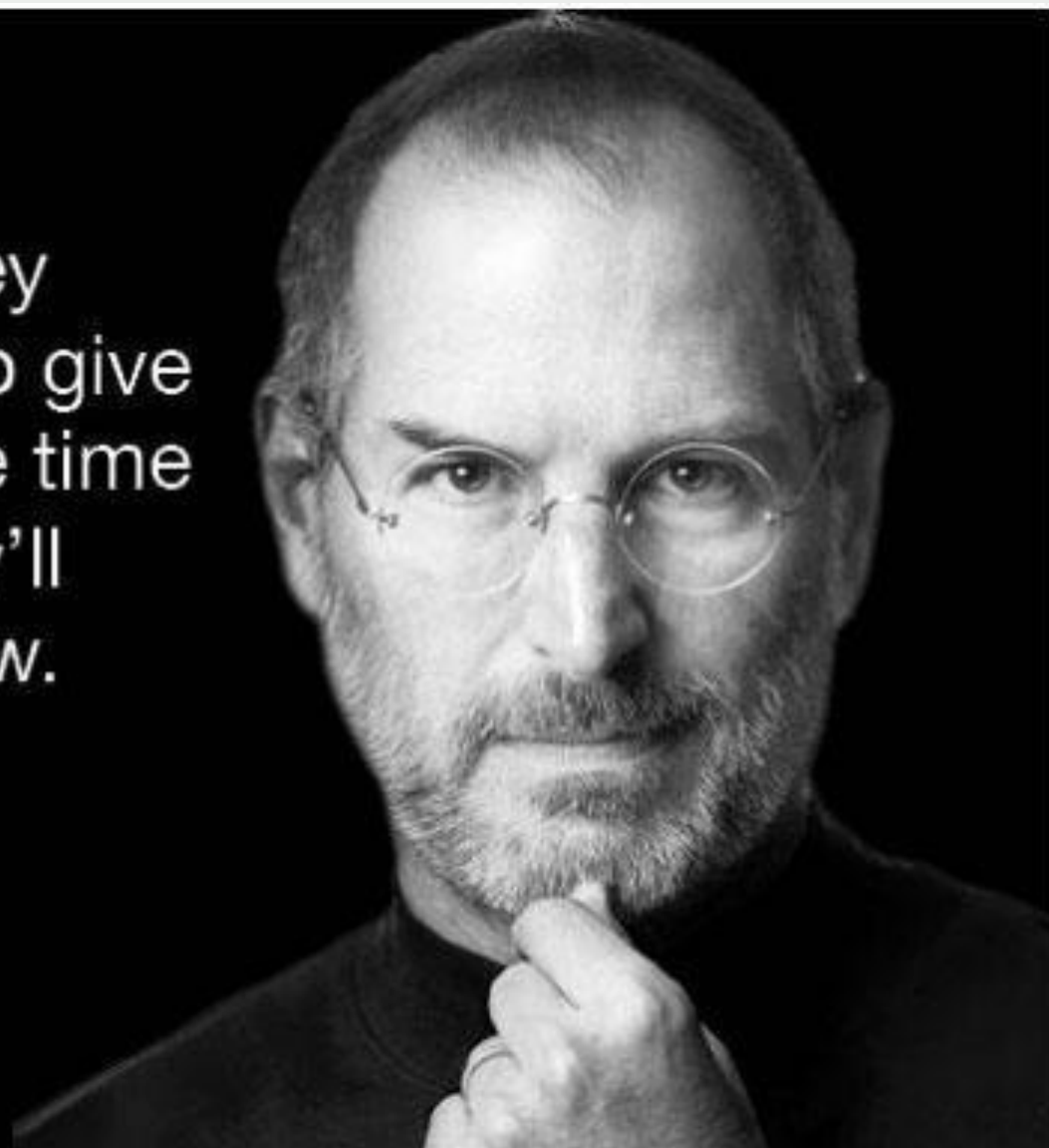
And always remember...

... YOU ARE NOT THE USER!



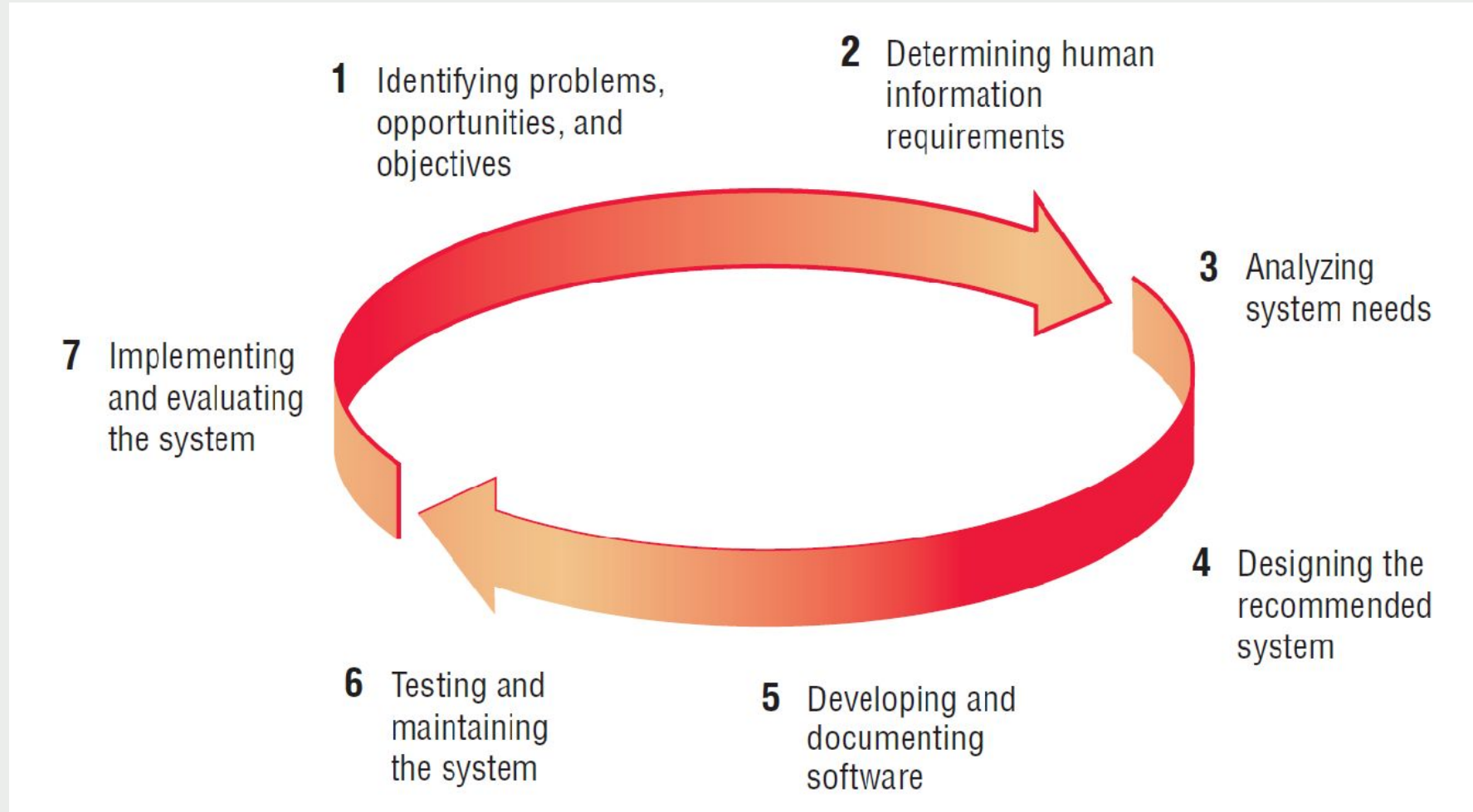
You can't just ask
customers what they
want and then try to give
that to them. By the time
you get it built, they'll
want something new.

1989



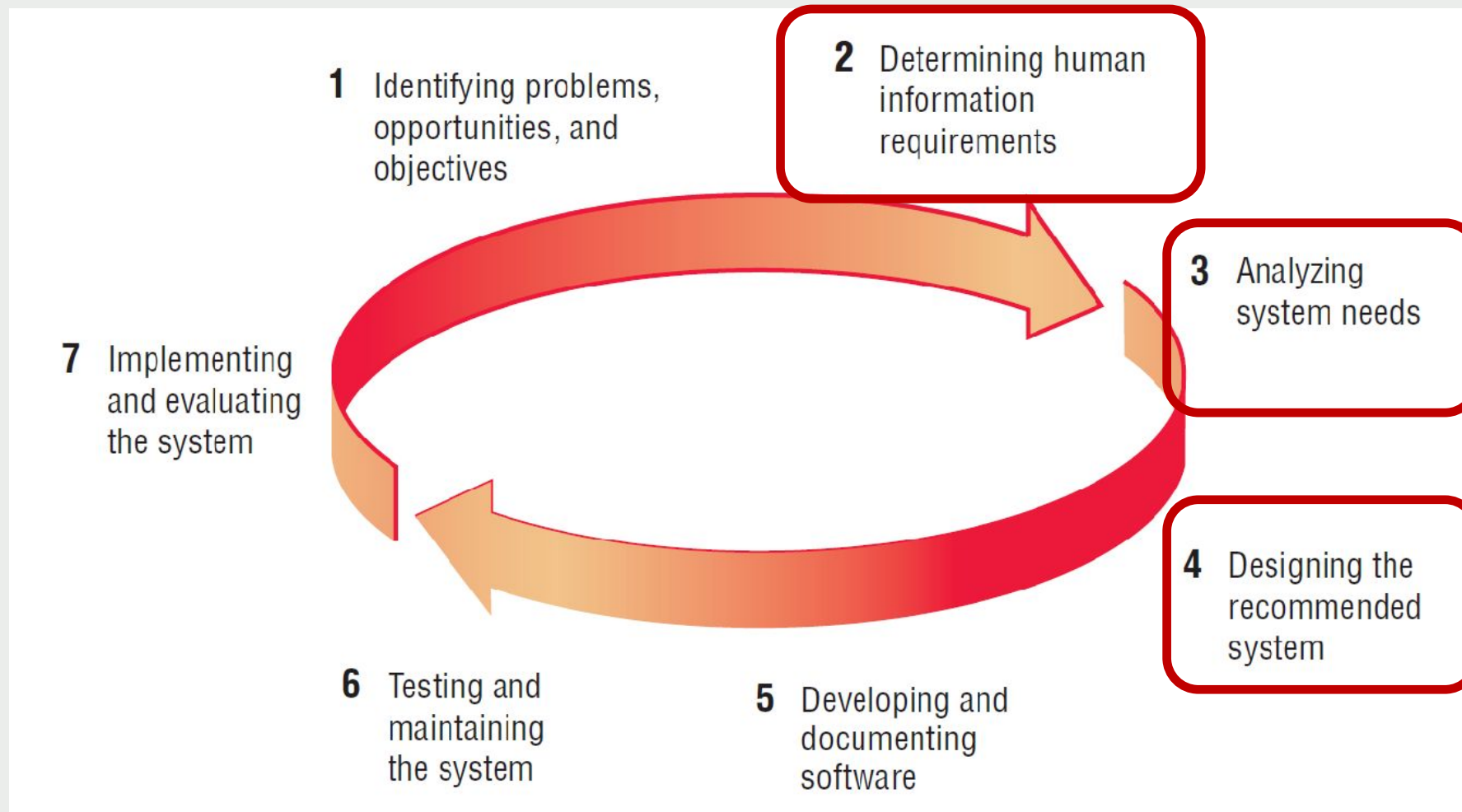
Systems Development Life Cycle (SDLC)

The systems development life cycle is a phased approach to solving business problems



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2 Determining human information requirements

Requirement Analysis (Functional & Non-Functional)

- What the system must do
- Performance needs
- Security, usability
- User & business needs

3 Analyzing system needs

Process Modeling

- Data Flow Diagrams (DFDs)
- Business Process Models (BPMN)
- Use-case diagrams

4 Designing the recommended system

System Modeling

- Class diagrams
- Sequence diagrams
- State diagram